pcbai value overlay assembly guide

Values shown directly on footprints, exact source precision (not rounded purchase values).

Each board side: full overview page, then 2x2 zoomed tiles (50 x 50 mm each) for printing.

DC1: four population variants a/b/c/d on identical bare PCB. Only fitted parts get values; DNP dimmed.

DC3: DNP sites dimmed. Back sides are MIRRORED (as seen when you flip the board over).

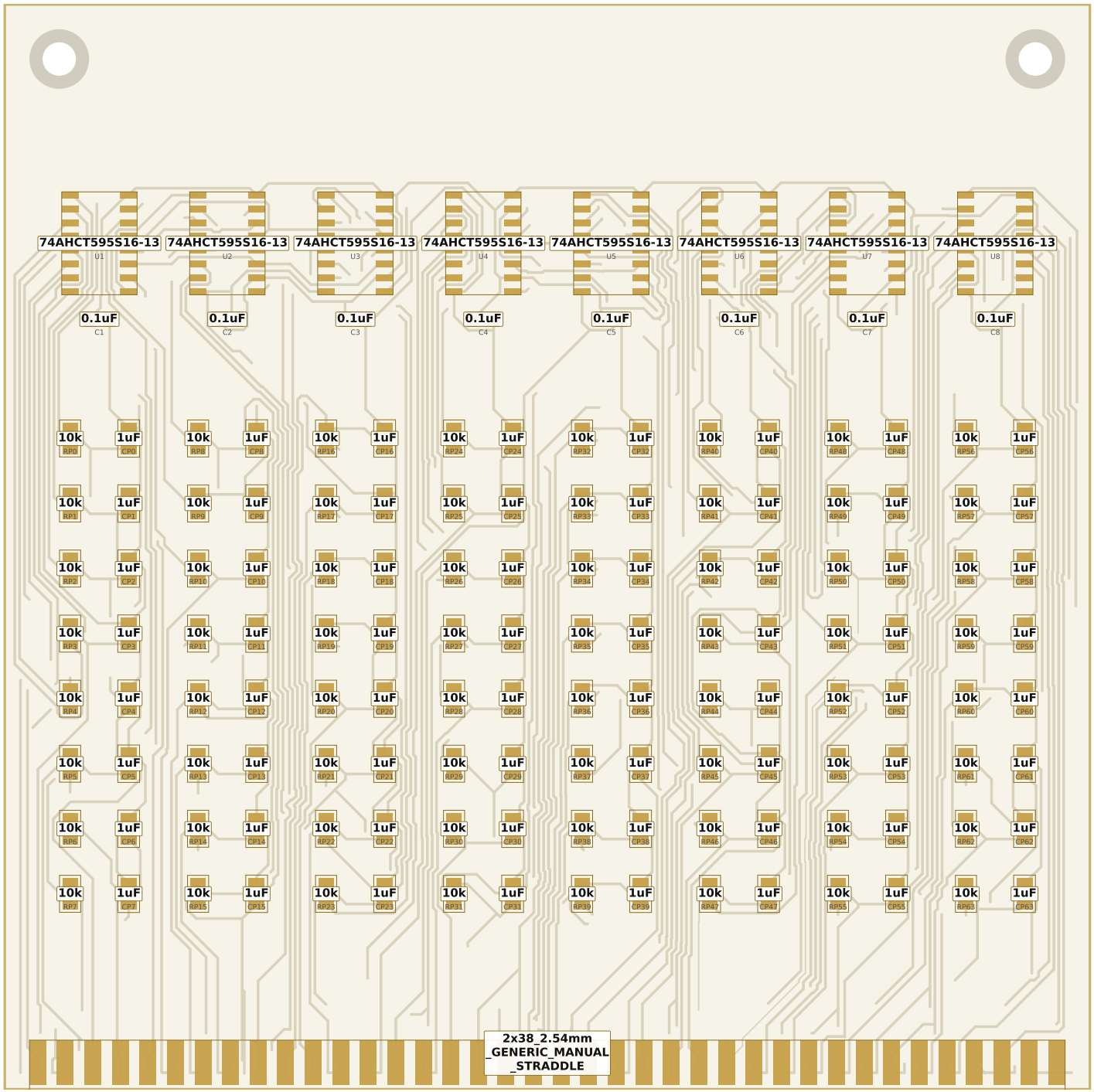
Motherboard: MANDATORY red bodge wire JS1 B7 (18.25, 44.27) to JS2 B7 (18.25, 55.27), back side.

DC0: source value 74AHCT595S16-13; the part actually ordered is 74HCT595 (5V-tolerant HCT, works as drop-in).

Source hashes: DC0_v2 899da15d, DC1 b5d0b202, DC2 04167d23, DC3_v2 d7c2393f, motherboard_v2 4932b44c

DC0 v2 TOP overview

source sha 899da15d | 145 fitted | board 100.05 x 100.05 mm

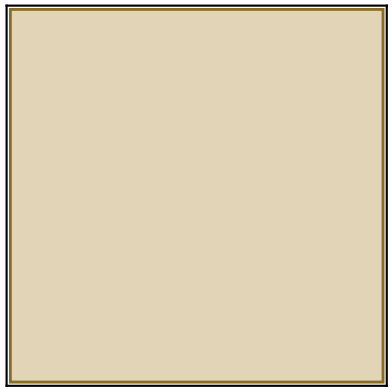


TOP (F)
viewed from above

- Fitted pad, bold value label
- DNP: dim + grey X, leave empty
- THT from other side / no part

Small grey text = reference.
Red leader = label moved off part.
Coords exact from KiCad source (mm).

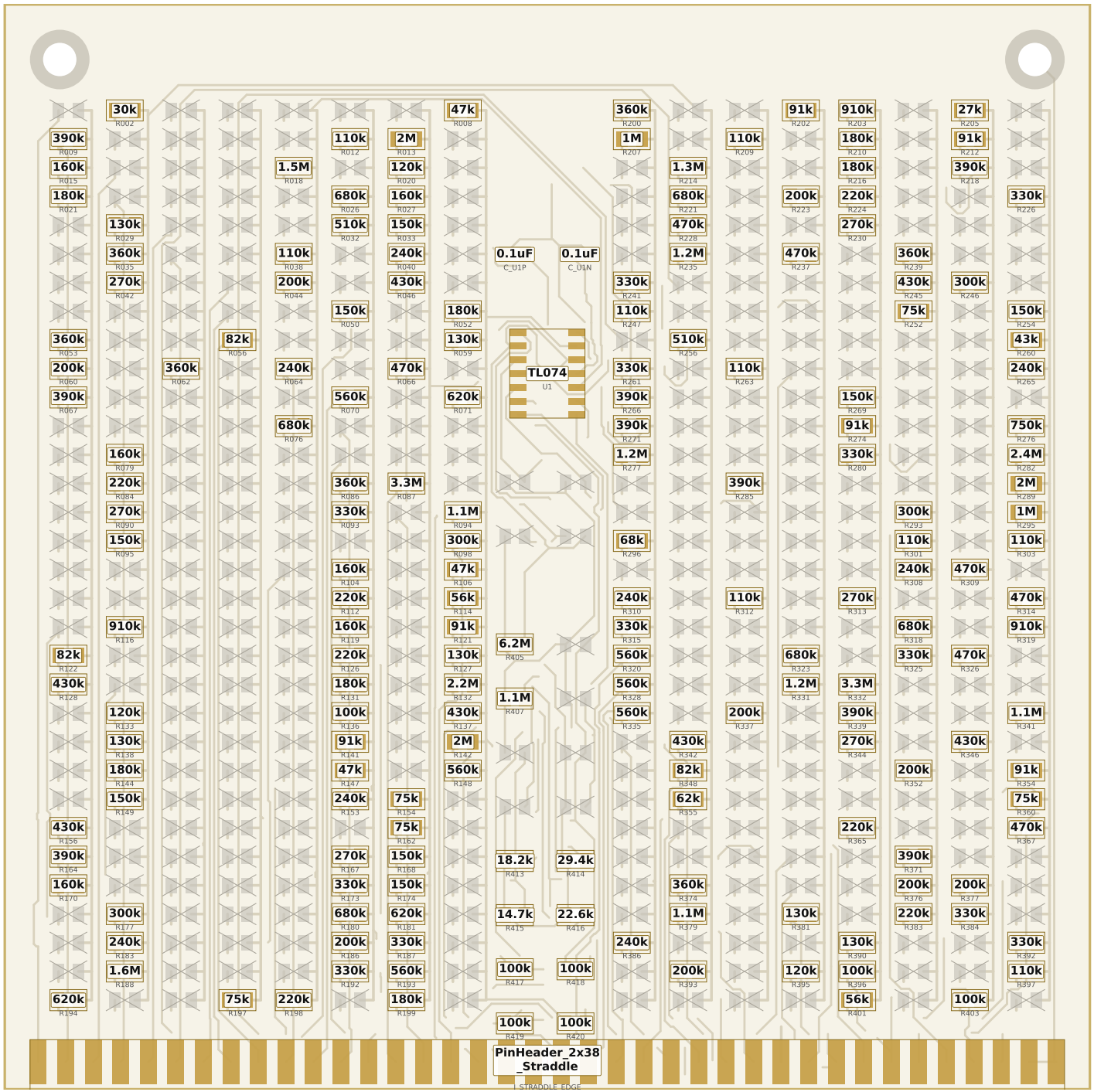
tile location



Overview, for orientation; use tiles to read.

DC1 variant a TOP overview

source sha b5d0b202 | 195 fitted, 341 DNP | board 100.15 x 100.15 mm

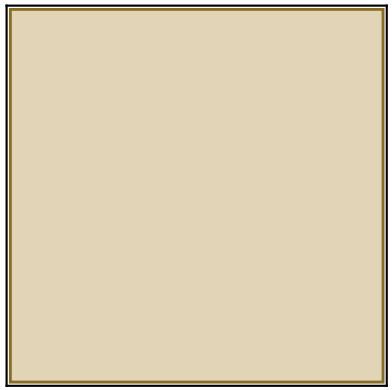


TOP (F)
viewed from above

- Fitted pad, bold value label
- DNP: dim + grey X, leave empty
- THT from other side / no part

Small grey text = reference.
Red leader = label moved off part.
Coords exact from KiCad source (mm).

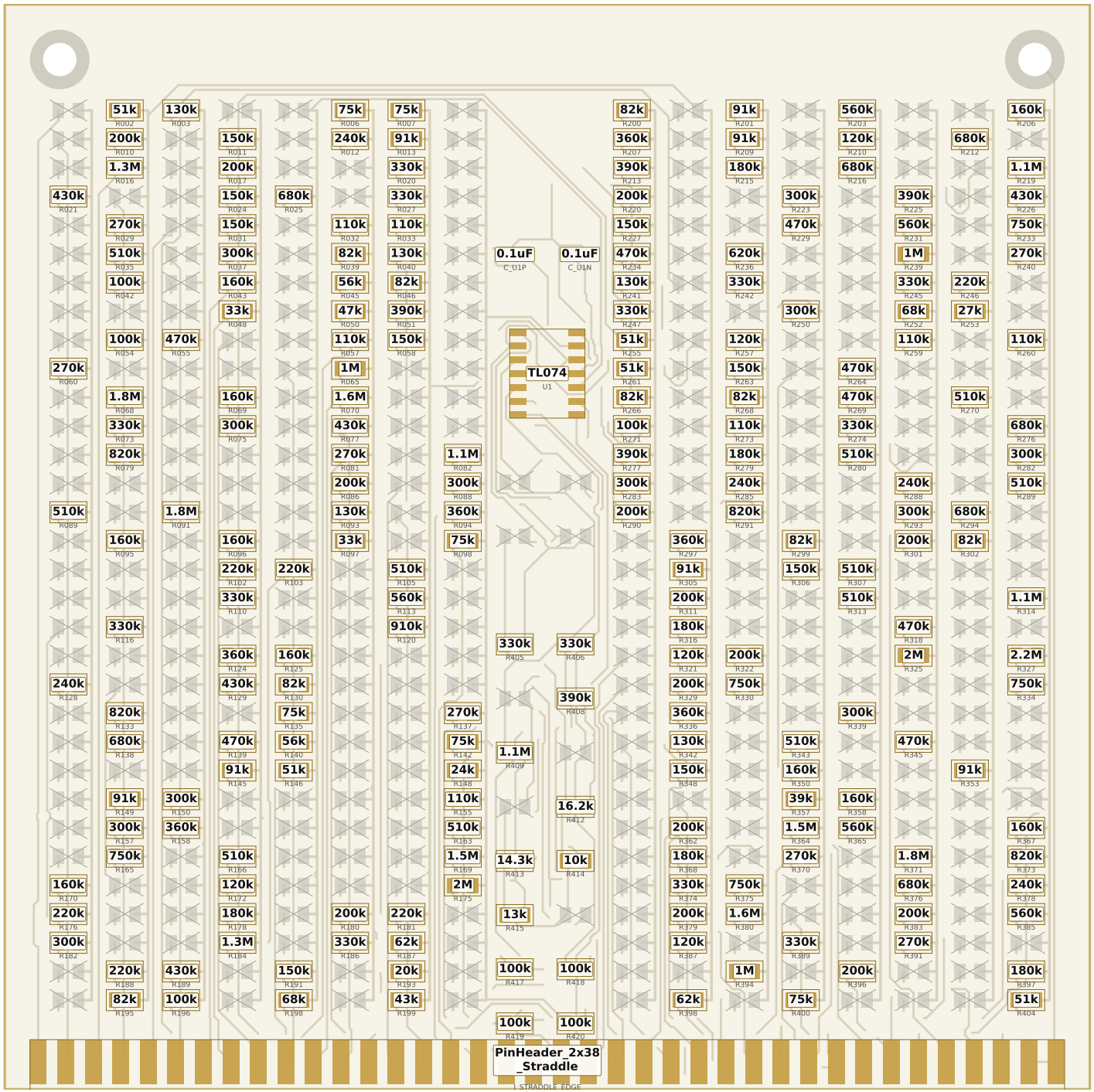
tile location



Overview, for orientation; use tiles to read.

DC1 variant b TOP overview

source sha b5d0b202 | 234 fitted, 302 DNP | board 100.15 x 100.15 mm

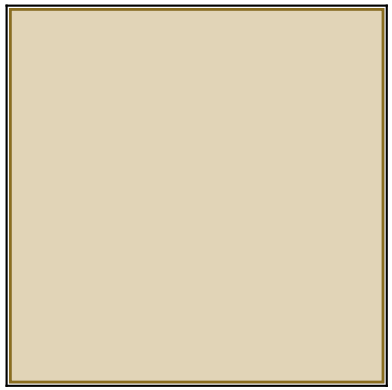


TOP (F)
viewed from above

- Fitted pad, bold value label
- DNP: dim + grey X, leave empty
- THT from other side / no part

Small grey text = reference.
Red leader = label moved off part.
Coords exact from KiCad source (mm).

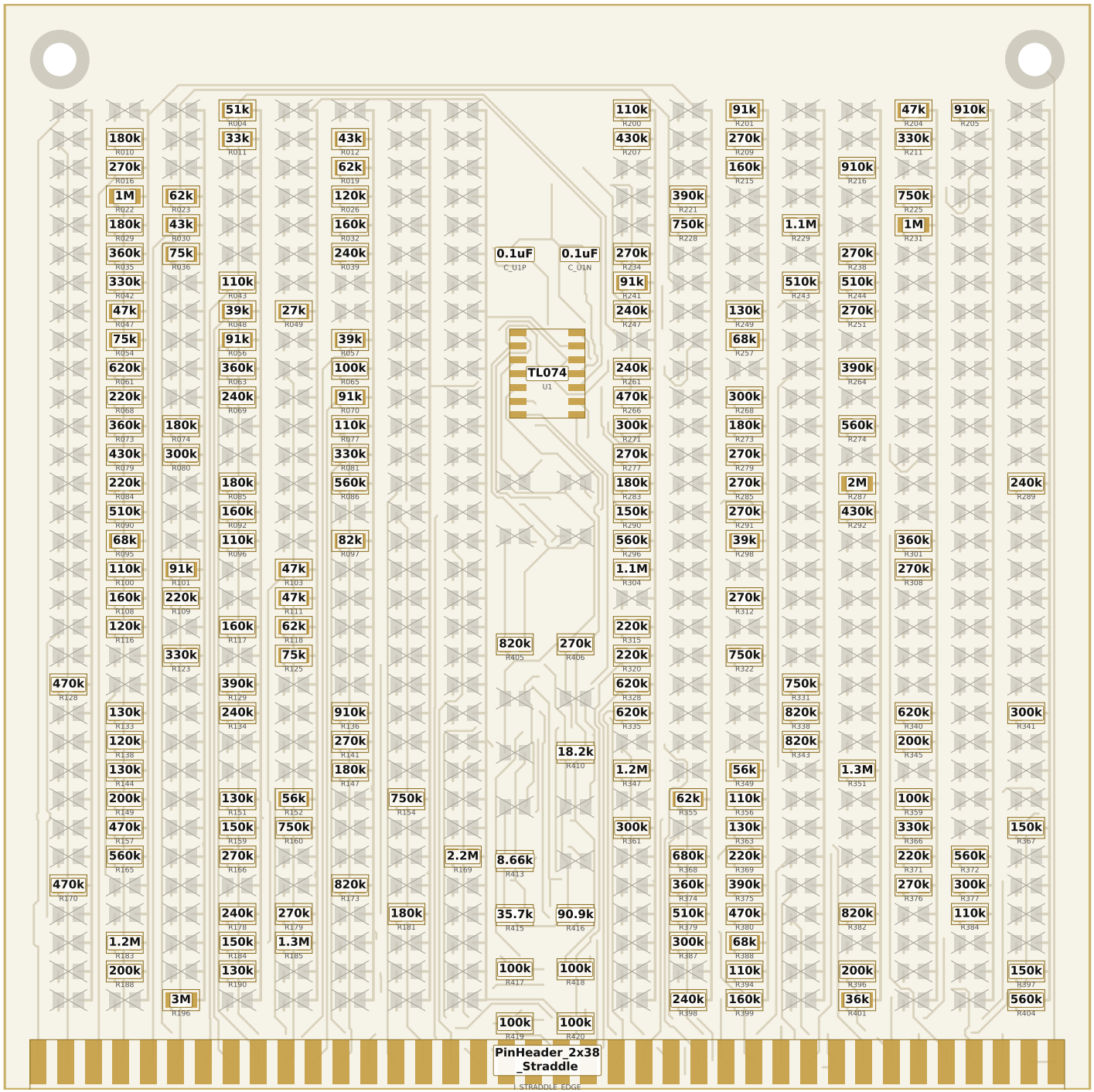
tile location



Overview, for orientation; use tiles to read.

DC1 variant c TOP overview

source sha b5d0b202 | 185 fitted, 351 DNP | board 100.15 x 100.15 mm

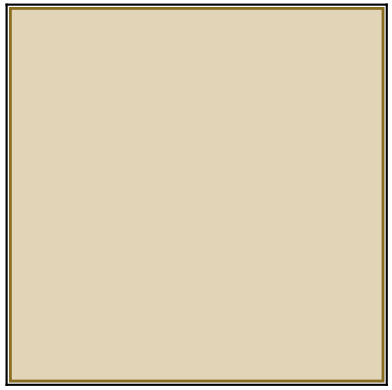


TOP (F)
viewed from above

- Fitted pad, bold value label
- DNP: dim + grey X, leave empty
- THT from other side / no part

Small grey text = reference.
Red leader = label moved off part.
Coords exact from KiCad source (mm).

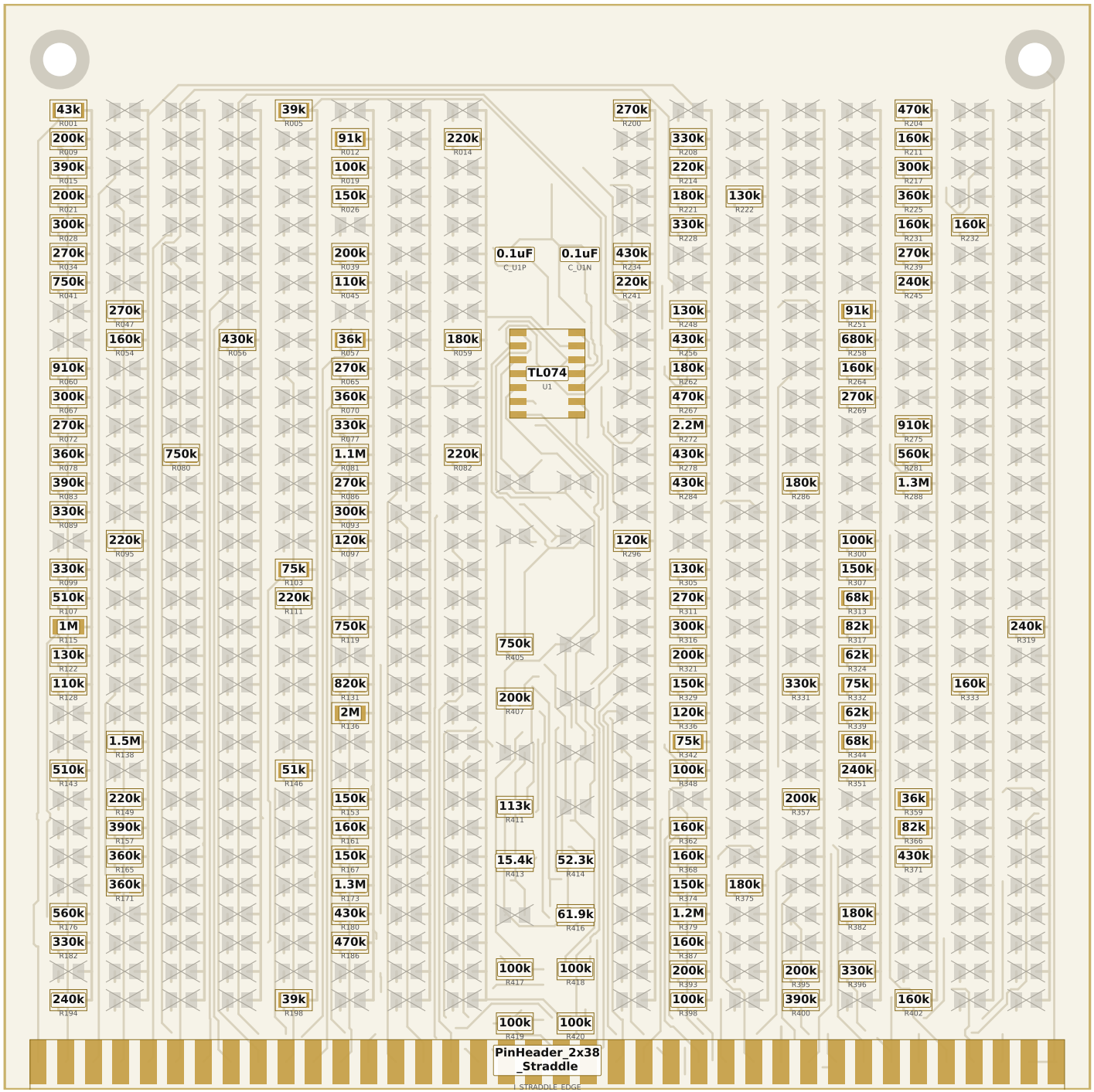
tile location



Overview, for orientation; use tiles to read.

DC1 variant d TOP overview

source sha b5d0b202 | 145 fitted, 391 DNP | board 100.15 x 100.15 mm

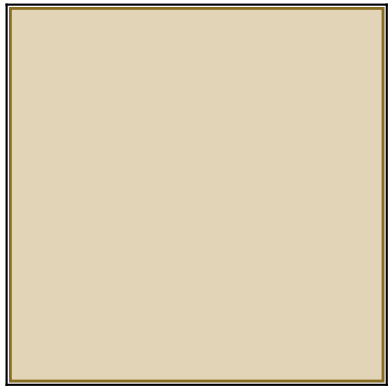


TOP (F)
viewed from above

- Fitted pad, bold value label
- DNP: dim + grey X, leave empty
- THT from other side / no part

Small grey text = reference.
Red leader = label moved off part.
Coords exact from KiCad source (mm).

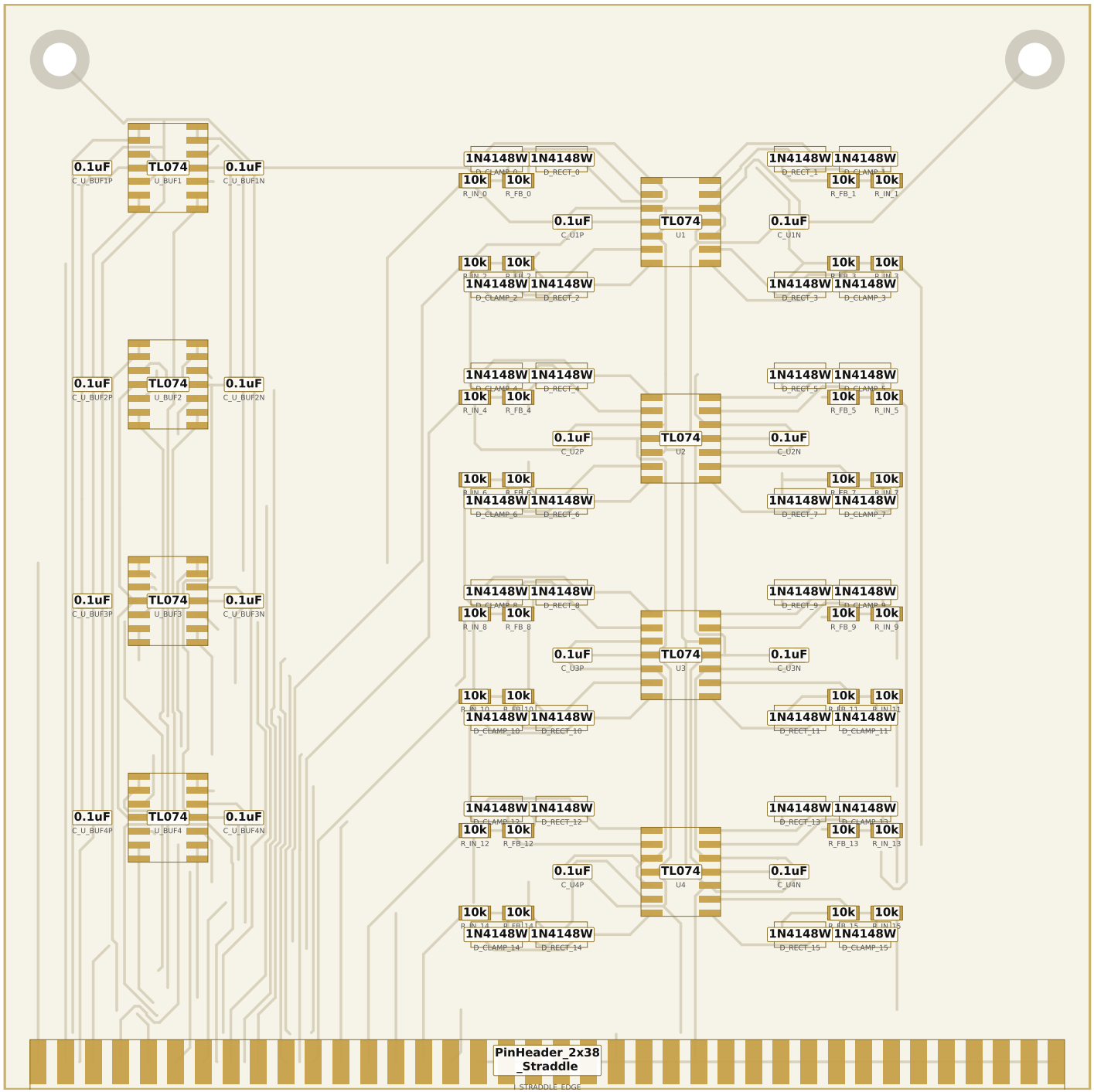
tile location



Overview, for orientation; use tiles to read.

DC2 TOP overview

source sha 04167d23 | 89 fitted | board 100.15 x 100.15 mm

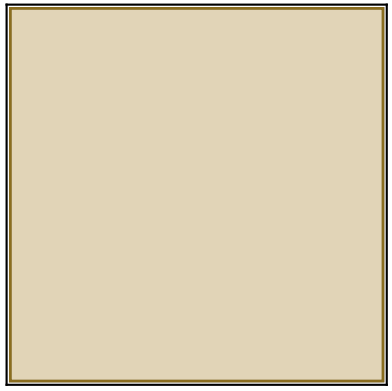


TOP (F)
viewed from above

- Fitted pad, bold value label
- DNP: dim + grey X, leave empty
- THT from other side / no part

Small grey text = reference.
Red leader = label moved off part.
Coords exact from KiCad source (mm).

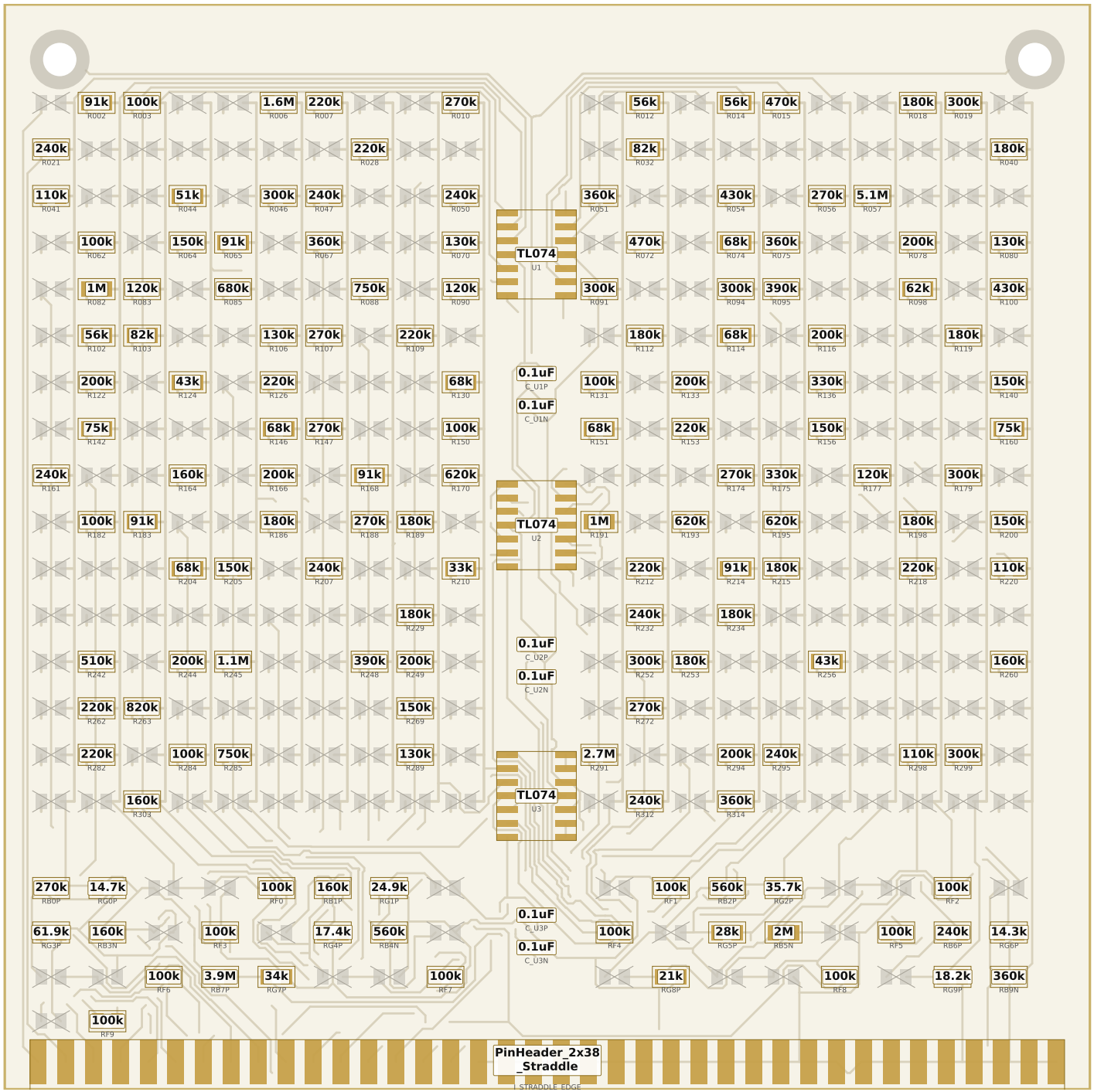
tile location



Overview, for orientation; use tiles to read.

DC3 v2 TOP overview

source sha d7c2393f | 163 fitted, 217 DNP | board 100.15 x 100.15 mm

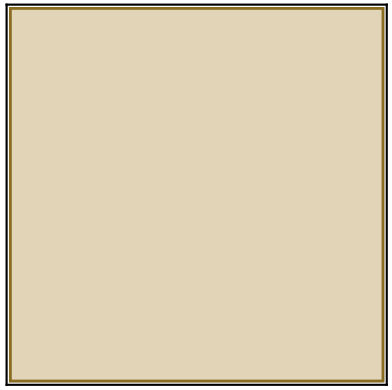


TOP (F)
viewed from above

- Fitted pad, bold value label
- DNP: dim + grey X, leave empty
- THT from other side / no part

Small grey text = reference.
Red leader = label moved off part.
Coords exact from KiCad source (mm).

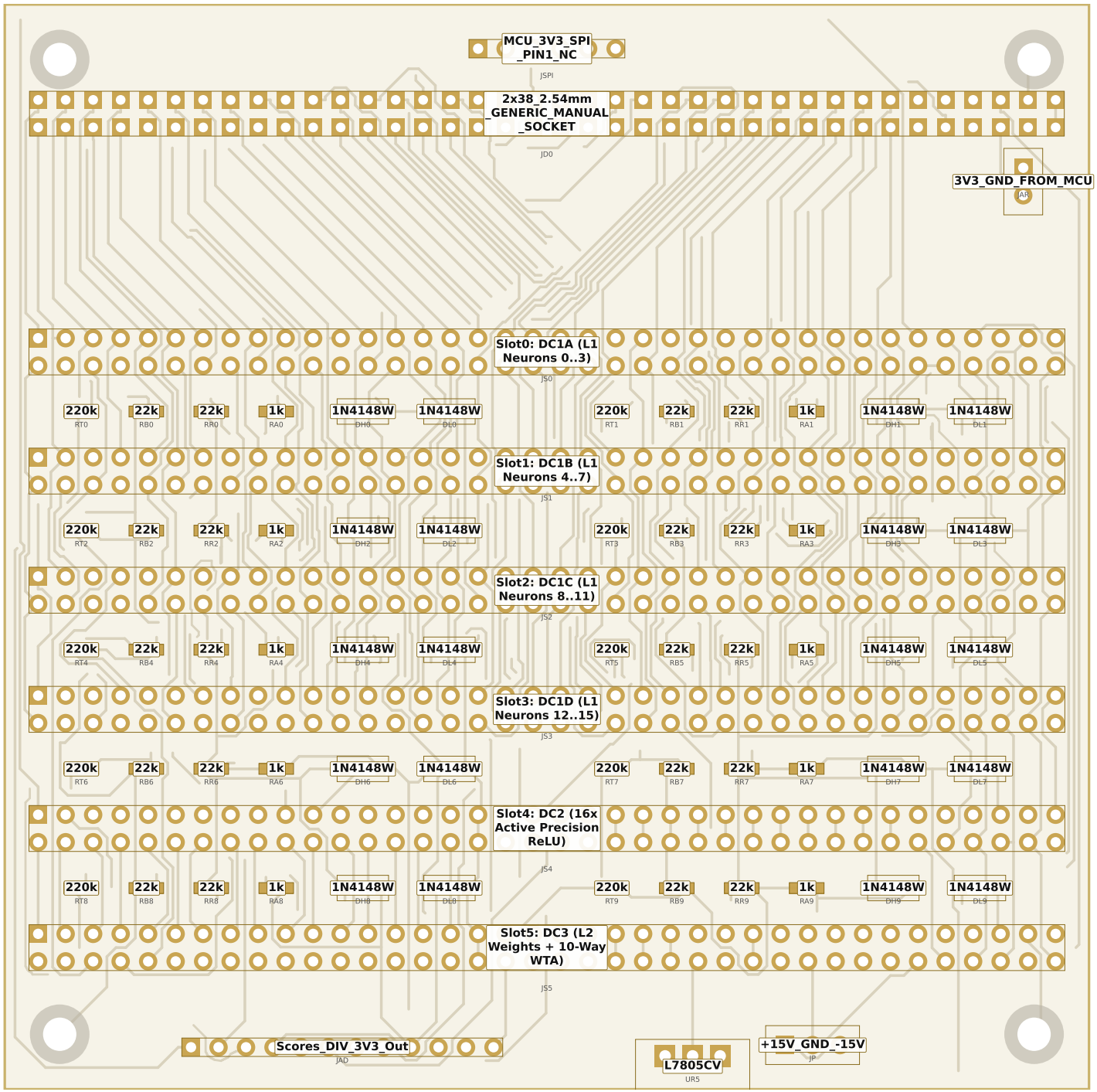
tile location



Overview, for orientation; use tiles to read.

Motherboard v2 TOP overview

source sha 4932b44c | 81 fitted | board 100.15 x 100.15 mm

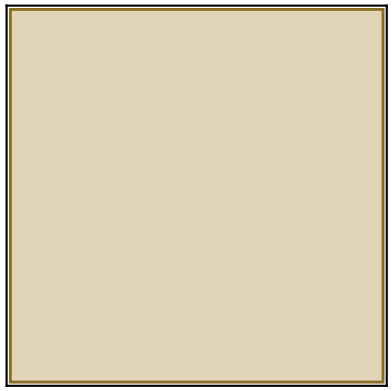


TOP (F)
viewed from above

- Fitted pad, bold value label
- DNP: dim + grey X, leave empty
- THT from other side / no part

Small grey text = reference.
Red leader = label moved off part.
Coords exact from KiCad source (mm).

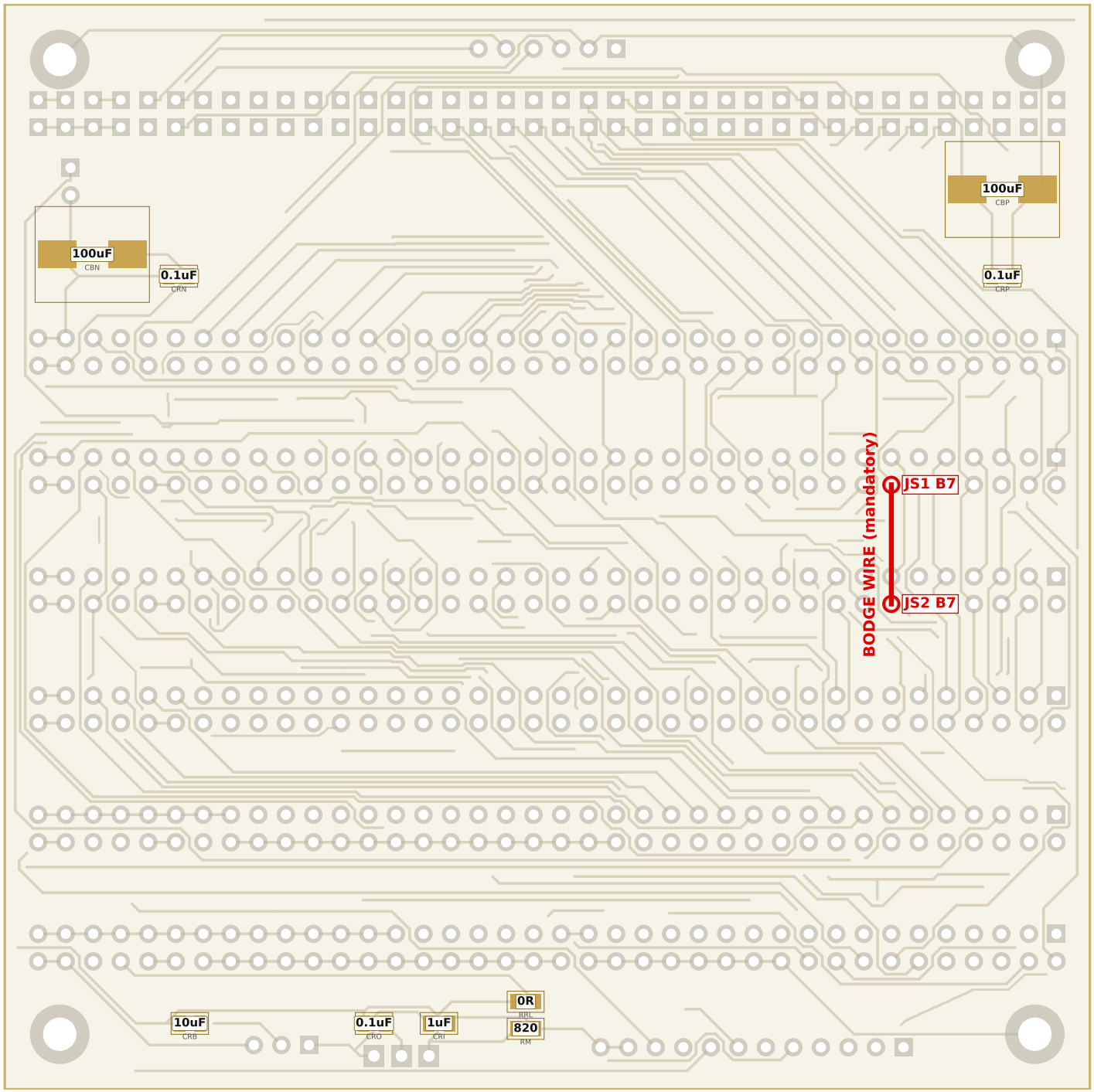
tile location



Overview, for orientation; use tiles to read.

Motherboard v2 BACK overview

source sha 4932b44c | 81 fitted | board 100.15 x 100.15 mm

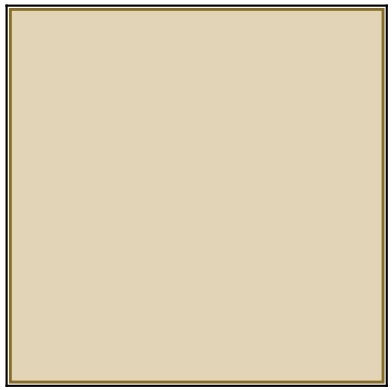


BACK (B), MIRRORED
viewed from below

- Fitted pad, bold value label
- DNP: dim + grey X, leave empty
- THT from other side / no part

Small grey text = reference.
Red leader = label moved off part.
Coords exact from KiCad source (mm).

tile location (mirrored)



Overview, for orientation; use tiles to read.